

Limits

ANSWER KEY



Direct substitution

- Find $\lim_{x \rightarrow 3} (2x^2 - x + 1)$.
 $= 2(9) - 3 + 1 = 16$.
- Find $\lim_{x \rightarrow -1} (x^3 + 4x)$.
 $= (-1)^3 + 4(-1) = -1 - 4 = -5$.

Indeterminate forms: factor and cancel

- Find $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$.
Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 16}{x - 4} = \frac{(x - 4)(x + 4)}{x - 4} = x + 4$ (for $x \neq 4$), so the limit is $4 + 4 = 8$.
- Find $\lim_{x \rightarrow -2} \frac{x^2 + 5x + 6}{x + 2}$.
Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + 5x + 6}{x + 2} = \frac{(x + 2)(x + 3)}{x + 2} = x + 3$ (for $x \neq -2$), so the limit is $-2 + 3 = 1$.
- Explain why you cannot find $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$ by direct substitution alone.
Substituting $x = 4$ gives $\frac{16 - 16}{4 - 4} = \frac{0}{0}$, an indeterminate form — it does not tell you the limit is 0 or undefined, so you must factor and cancel first.

Limits at infinity for rational functions

- Find $\lim_{x \rightarrow \infty} \frac{3x^2 + 2x}{5x^2 - 1}$.
The numerator and denominator have the same degree (2), so by Theorem 6.1 the limit is the ratio of leading coefficients: $\frac{3}{5}$.
- Find $\lim_{x \rightarrow \infty} \frac{4x + 1}{2x^2 + 3}$.
The denominator has higher degree than the numerator, so by Theorem 6.1 the limit is 0.
- Explain what happens to $\lim_{x \rightarrow \infty} \frac{5x^3 - 2}{x^2 + 1}$, without computing an exact value.
The numerator's degree (3) is higher than the denominator's (2), so by dividing by the highest power in the denominator the numerator's leading term still grows without bound — the limit does not exist as a finite number; the function increases without bound as $x \rightarrow \infty$.